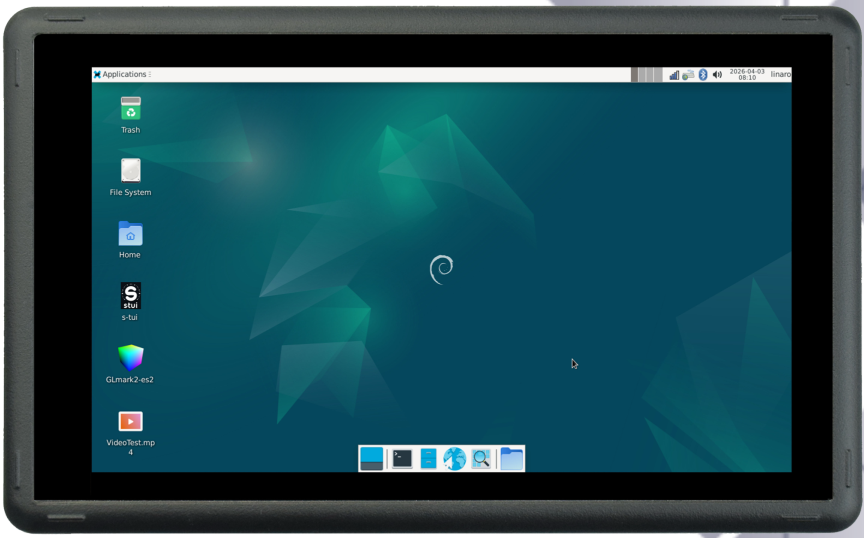
Industrial PC

PPC-RK3576-050



PN: CS12720-RK3576-050P

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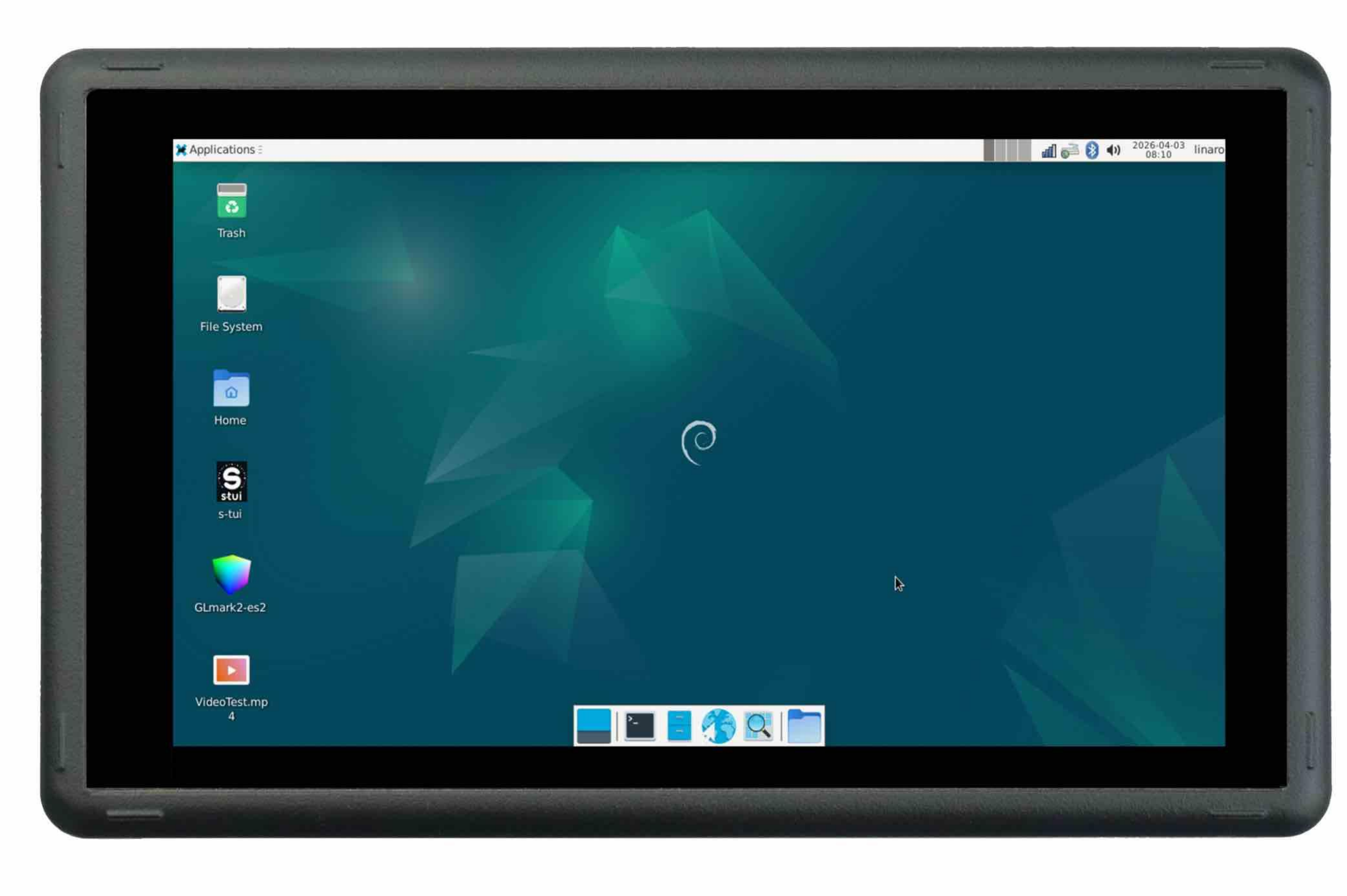
Contents

PPC-RK3576-050

Front View



Rear View


LAN
USB HOST
Chipsee
Type:CS12720-RK3576-050P
Rev:C111
OS: Debian 12
SN:2026040310003757
https://chipsee.com
Origin:Made in China
0 697417 431079

Side View 1


CAN1_H
CAN1_L
RS485_5-
RS485_5+
RS485_3-
RS485_3+
RS232_2_RDX
RS232_2_TXD
RS232_0_RXD
RS232_0_TXD
GND
+5V OUT
G
-
+
DC:9~30V
Applications
Trash
File System
Home
GLmark2-es2

Side View 2

Product Overview

The Cortex®-A72+ Cortex®-A53 series PPC-RK3576-050 (PN: CS12720-RK3576-050P) is a high-quality IP65-compliant industrial panel PC. This single board computer features a 5.0” 5-point capacitive touch screen with 1mm Armored Glass with a resolution of 1280 x 720 pixels and a brightness of 400 cd/m^2.

Key Applications

- Human Machine Interface HMI
- Mobile Applications
- Video Processing
- Machine Learning
- Video Gaming
- Process Control
- Process Monitoring
- ATM…

It is available as a device hosed in an aluminum casing with bezels, thus facilitating different installation options:

- Installation on an industrial cabinet
- Integration with the existing equipment

The PPC-RK3576-050 Industrial Panel PC is based around the powerful RK3576 System on Chip (SoC), powered by the Rockchip RK3576 low-power processor which integrates a Quad-core Cortex-A72(1.6GHz) + Quad-core Cortex-A53(1.4GHz) processor, and a 6 TOPS(Sparsity)@INT8 computing power NPU (neural processing unit).

The RK3576 supports multi-format video decoders and has a high-performance 4GB LPDDR5 RAM capable of sustaining demanding memory bandwidth. It also provides a complete set of peripheral interfaces.

Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order the PPC-RK3576-050 from the official Chipsee Store or from your nearest distributor.

Operating System

This product comes with a pre-installed OS of your choice. Please see the list below for the supported OSes, which can also be obtained from the Software Documentation section, along with the detailed installation instructions.

- Android 14
- Debian 12
- Buildroot Linux Qt 5.15

> **Warning**
>
> The Software Documentation section provides a detailed instruction on how to install different OSes on your own. However, bear in mind that Chipsee can't take the responsibility of inadequate installation procedure. If you "brick" your device, please contact Chipsee Technical Support at **support@chipsee.com** for further assistance

Hardware Features

The PPC-RK3576-050 Industrial Panel PC offers a broad range of performance and connectivity options for scalable integration, providing expandability according to future needs. Some of the key features are listed in the table below.

PPC-RK3576-050	
CPU	Rockchip RK3576J, Quad-core Cortex-A72 (1.6GHz) + Quad-core Cortex-A53 (1.4GHz)
RAM	4GB LPDDR5
NPU	6 TOPS(Sparsity)@INT8
eMMC	64GB
Storage	TF Card, Supports up to 128GB SDHC
PCIe	N/A
Display	5.0” LCD, 1280 x 720, High Brightness: 400 cd/m^2
Touch	5-point capacitive touch screen with 1mm Armored Glass
HDMI	1 x HDMI-D (Micro-HDMI) Out
USB	1 x USB 2.0 HOST, 1 x USB 3.0 HOST, 1 x USB Type-C (USB 3.0 port and USB-C port cannot be used at the same time)
LAN	1 x RJ45, GbE
POE	N/A
Audio	3.5mm Audio In/Out Connector, 2W Internal Speaker
Buzzer	Yes
RTC	High accuracy RTC with farad capacitor, can work 1 week after power off **(default)**. High accuracy RTC with lithium coin battery, can work 3 years after power off *(optional)*.
RS232	Default to 2 x RS232 (including 1 debug port). Up to 4 x RS232 by swapping 2 x RS485 to RS232.[1]
RS485	Default to 2 x RS485. Optionally, these 2 x RS485 can be configured to RS232. Optionally, 1 x RS485 can be configured to CAN(CAN1)[1]
CAN	Default to 1 x CAN FD. Optionally, 1 x RS485 can be configured to CAN(CAN1), providing up to 2 x CAN FD.
GPIO	N/A
WiFi/BT	Integrated WiFi/Bluetooth Module
4G/LTE	N/A
Power Input	From 9V to 30V
Current	450mA Max at 15V

PPC-RK3576-050	
Power Consumption	5.4W Max
Working Temperature	From -20°C to +60°C
OS	Android 14, Debian12, Buildroot Linux Qt 5.15
Dimensions	PPC-RK3576-050 (PN: CS12720-RK3576-050P): 138.55 x 84.70 x 27.10mm
Weight	PPC-RK3576-050 (PN: CS12720-RK3576-050P): 310g
Mounting	PPC-RK3576-050 (PN: CS12720-RK3576-050P): Panel, VESA

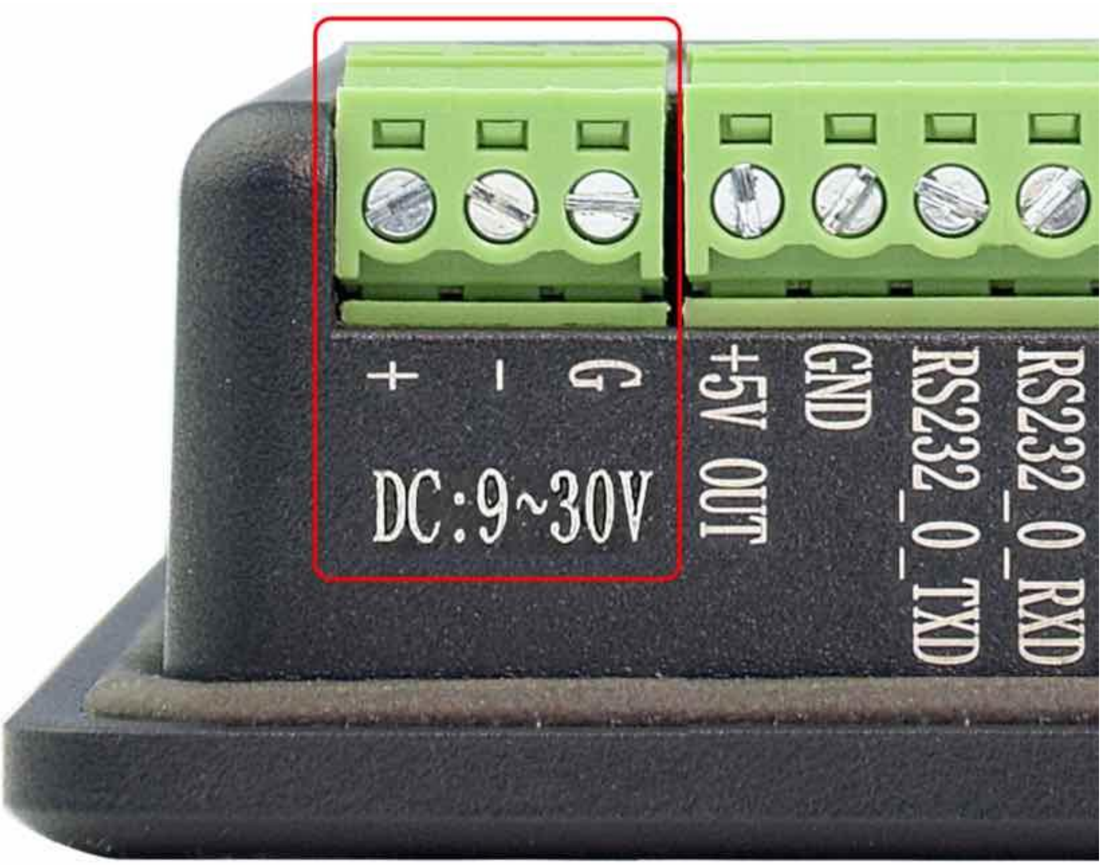
Key Features

1(1,2) This product has 4 x UART by default. The default configuration is 2 x RS232 and 2 x RS485, including 1 RS232 debug port. The 2 x RS485 can be configured to 2 x RS232. UART can be swapped between RS232 and RS485 modes easily, if you need a different RS232/RS485 configuration, please get in touch with the Chipsee Technical Support at **support@chipsee.com**

Power Input

The PPC-RK3576-050 Industrial Panel PC can be powered by a wide range of input voltages: From 9V to 30V DC.

There is a DC input interface on this device: a **3-pin, 3.81mm screw terminal** connector. As shown in the figure below.



Power Input

Note that the "**+**" sign represents the positive power input. The "**-**" terminal is shorted to the ground.

Power Input Definition		
Pin Number	**Definition**	**Description**
Pin 1	Positive Input	DC Power **Positive Terminal**
Pin 2	Negative Input	DC Power **Negative Terminal**
Pin 3	Ground	**Power System Ground**

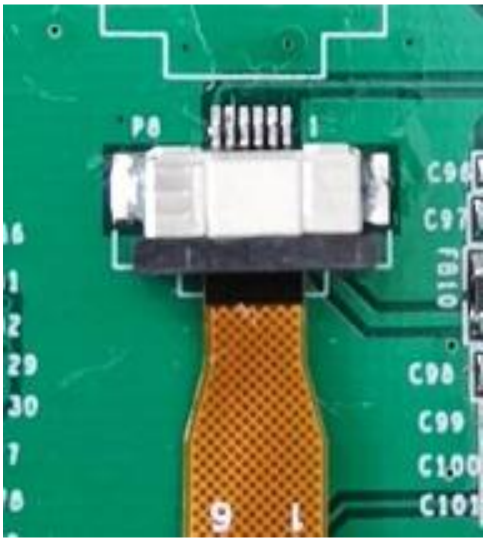
Power Connector

Note

> The system ground “**G**” is connected to power negative “**-**” on board.

Touch Screen

The PPC-RK3576-050 Industrial Panel PC uses a 5-point capacitive touch screen with 1mm Armored Glass. However, the Debian OS supports only One-Point touch.



Capacitive Touch Connector

Attention

A capacitive touch screen is susceptible to power noise and Electromagnetic Radiation (EMR). It may cause LCD ripples or even capacitive touch malfunction. If using a capacitive multi-touch test application, you might notice the touch points float erratically across the display. There are several solutions to this problem:

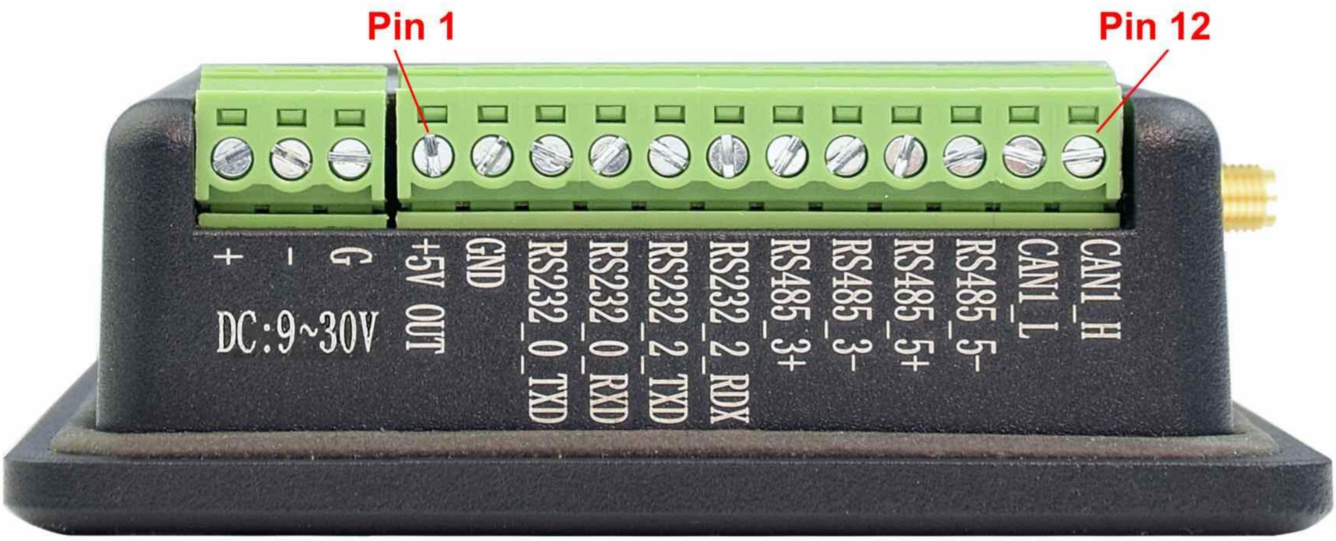
1. Use a high-quality Power Adapter Unit (PSU) with low EMR. You can also provide power from a battery.
2. Make sure that the PPC-RK3576-050 Power Input connector (pin 3) is properly connected to the Power System Ground to provide sufficient EMI shielding and eliminate the problem entirely.
3. Bad GND problem can also be confirmed by touching pin 3 of the Power Input connector with one hand while operating the capacitive touch screen with the other hand. In this case, the operator's body acts as the Power System Ground.

Connectivity

There are many connectivity options available on the PPC-RK3576-050 industrial PC. It has 1 x USB 2.0 HOST, 1 x USB 3.0 HOST, 1 x USB Type-C, 1 x network connector (RJ45) supporting up to 1 Gbps, 1 x CAN FD and 4 x UART terminals (RS232/485).

RS232+RS485+CAN

The serial communication interfaces (RS485, RS232, and CAN FD) are routed to a **12-pin 3.81mm terminal**, as illustrated in the figure below. Serial communication on both RS485 and RS232 interfaces can reach up to 115200 kbps.



Attention

- The 120Ω match resistor for **CAN** bus is **NOT mounted** by default.
- The 120Ω match resistor for **RS485** is **already mounted** by default.
- This product supports changing 2 x RS485 to 2 x RS232, providing up to 4 x RS232 (including one debug port).
- This product supports changing 1 x RS485 to 1 x CAN, providing up to 2 x CAN FD. In this case, Pin 10 is CAN0 H, Pin 9 is CAN0 L, OS node is CAN0.

The table below offers a detailed description of every pin:

Pin Number	Definition	Description	OS Node
Pin 12	CAN1_H	CPU SPI0, CAN H signal	
Pin 11	CAN1_L	CPU SPI0, CAN L signal	CAN1
Pin 10	RS485_5-	CPU UART5, RS485 –(B) signal	
Pin 9	RS485_5+	CPU UART5, RS485 +(A) signal	/dev/ttyS5
Pin 8	RS485_3-	CPU UART3, RS485 –(B) signal	

Pin Number	Definition	Description	OS Node
Pin 7	RS485_3+	CPU UART3, RS485 +(A) signal	/dev/ttyS3
Pin 6	RS232_2_RXD	CPU UART2, RS232 RXD signal	
Pin 5	RS232_2_TXD	CPU UART2, RS232 TXD signal	/dev/ttyS2
Pin 4	RS232_0_RXD	CPU UART0, RS232 RXD signal, Debug Port	
Pin 3	RS232_0_TXD	CPU UART0, RS232 TXD signal Debug Port	/dev/ttyFIQ0
Pin 2	GND	System Ground	
Pin 1	+5V	System +5V Power Output, No more than 1A Current output	

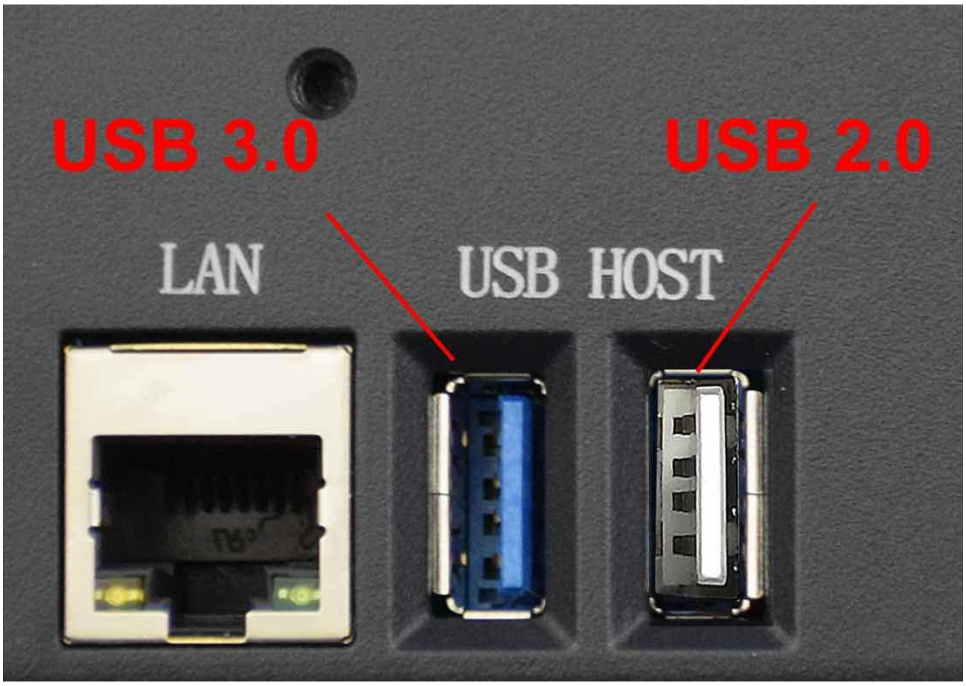
RS232 / RS485 / CAN Pin Definition for 5 inch product

USB Connectors

There are 2 x **USB HOST** and 1 x **USB DEVICE** (for flashing OS) ports onboard: 1 x USB 2.0 HOST, 1 x USB 3.0 HOST, 1 x USB Type-C, as shown in the figures below.

> **Warning**
>
> **Please unplug USB mouse/keyboard** from the USB3.0 port when flashing OS.

The USB 3.0 type-A host and USB-C **can't be used** at the same time. Before boot into OS, USB-C is enabled for installing OS image; after boot into OS, USB-A is enabled but USB-C is disabled.
In Android, these can be configured, USB3.0 type A is enabled by default, but you can switch to using USB-C in the software for debugging; in Linux, these can't be configured.



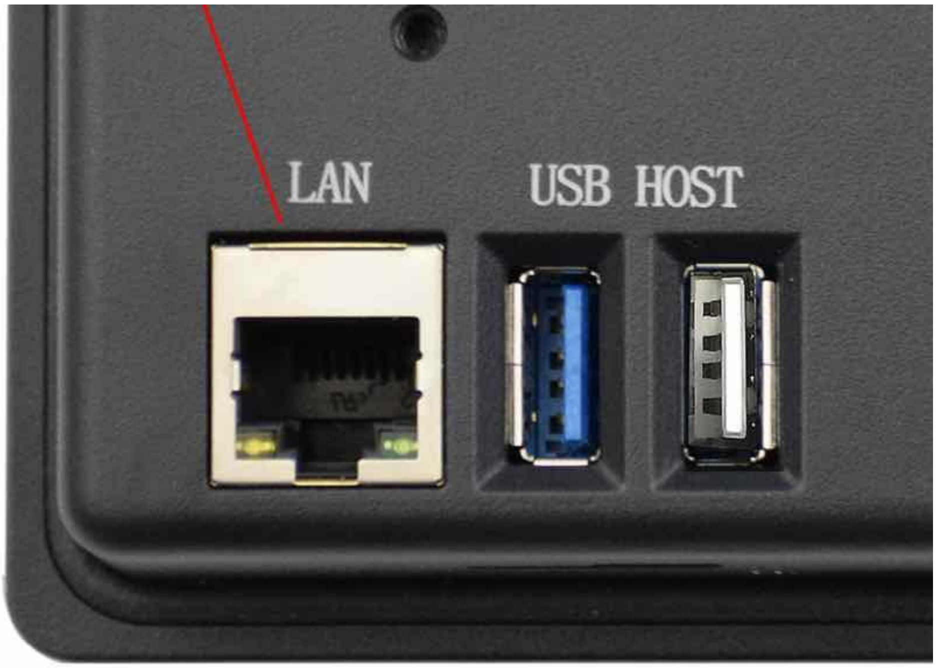
USB HOST Ports

USB Type-C Port

LAN Connector

LAN (RJ45) connector provides 1 x RJ45 Ethernet connectivity over standardized Ethernet cables as shown in the figure below. The integrated Ethernet interface supports up to 1 Gbps data throughput.

This product does not support PoE(Power over Ethernet).



RJ45 LAN Connector

> **Note**
>
> Use CAT5 or better cables to achieve full data throughput over maximum distance defined by the 1000BASE-T standard (100m).

WiFi & BlueTooth Module

The PPC-RK3576-050 Industrial Panel PC is equipped with the popular **Realtek RTL8821CS WiFi/BT module** which supports BT/BLE 2.1/3.0/4.2, as well as 802.11ac/abgn 433Mbps 2.4/5.8 GHz Wireless LAN (WLAN).

RTL8821CS WiFi/BT Module

The PPC-RK3576-050 includes an SMA connector for an external WiFi/BT antenna, as illustrated in the figure below.

WiFi+BT Antenna SMA

TF Card Slot

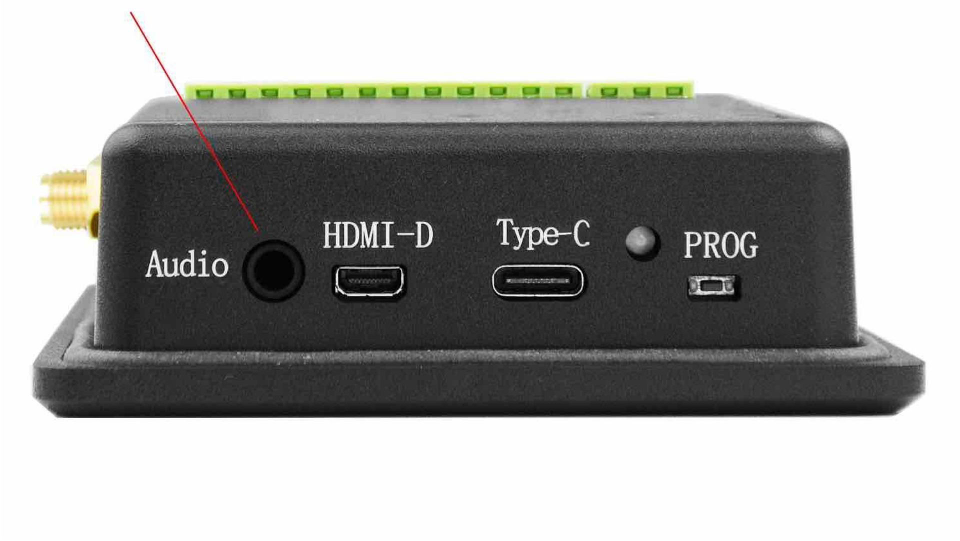
The PPC-RK3576-050 Industrial Panel PC features 1 x **TF Card (micro SD) slot**. TF Card can address up to 128GB of storage.

TF (micro SD) Card Slot

> **Note**
>
> The product does not come shipped with the TF Card by default.

Audio Connectors

The PPC-RK3576-050 Industrial Panel PC features some audio peripherals. It has a **3.5mm audio input/output jack**, an **internal speaker**, as well as a small **buzzer**.



Audio Jack

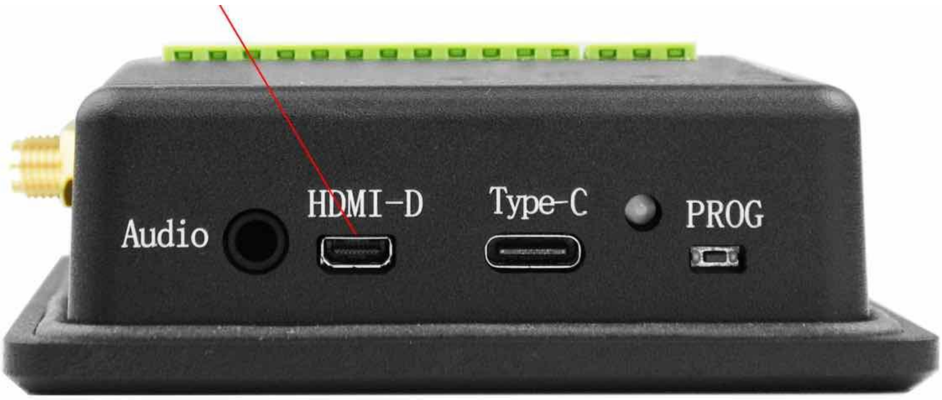
The miniature 2W embedded speaker is handy for audio reproduction, the small buzzer can play alarm/notification sounds.

2W Micro Speaker and Buzzer

> **Attention**
>
> By plugging in the headphone cable, the internal speaker will be disabled automatically.

HDMI Connector

The PPC-RK3576-050 Industrial Panel PC is equipped with 1 x HDMI-D (Micro-HDMI) Out port. The HDMI connector allows connecting an additional (external) monitor. HDMI output resolution can be configured by the software.



HDMI Connector

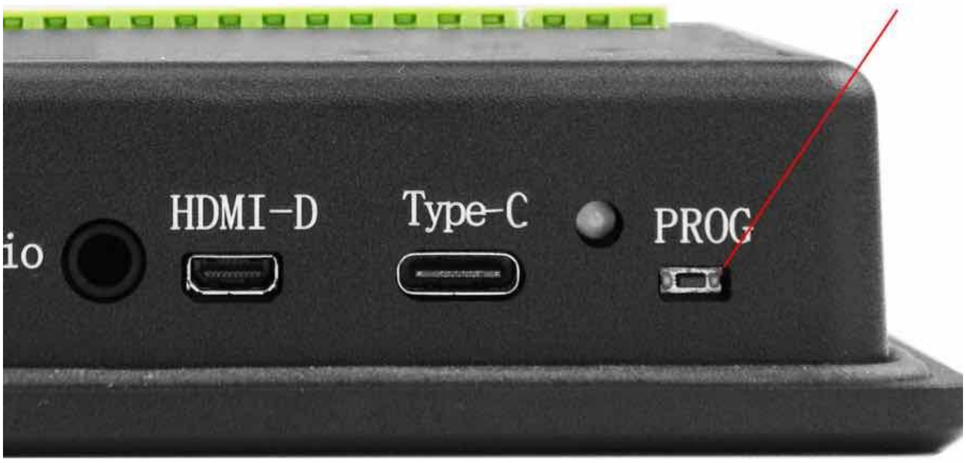
PROG Button

The PPC-RK3576-050 Industrial Panel PC has one button on the board marked as PROG, as shown in the figure below.

When the button is pressed before powering up, the PPC-RK3576-050 will enter LOADER mode. In this mode you can use a USB Type-C cable to upgrade its operating system. You can use this feature to flash another OS to the internal eMMC.

When the button is not pressed before and during power up, the PPC-RK3576-050 will boot normally.

There is no need to press the button during regular operation. However, if you need to flash the OS in MASKROM mode, the button will be used. Please refer to the software documents for more information.



PROG Button

Mounting Procedure

You can mount PPC-RK3576-050 with VESA mounting (guide): **75 x 75** mm, 4 x **M4** (6mm) screws.

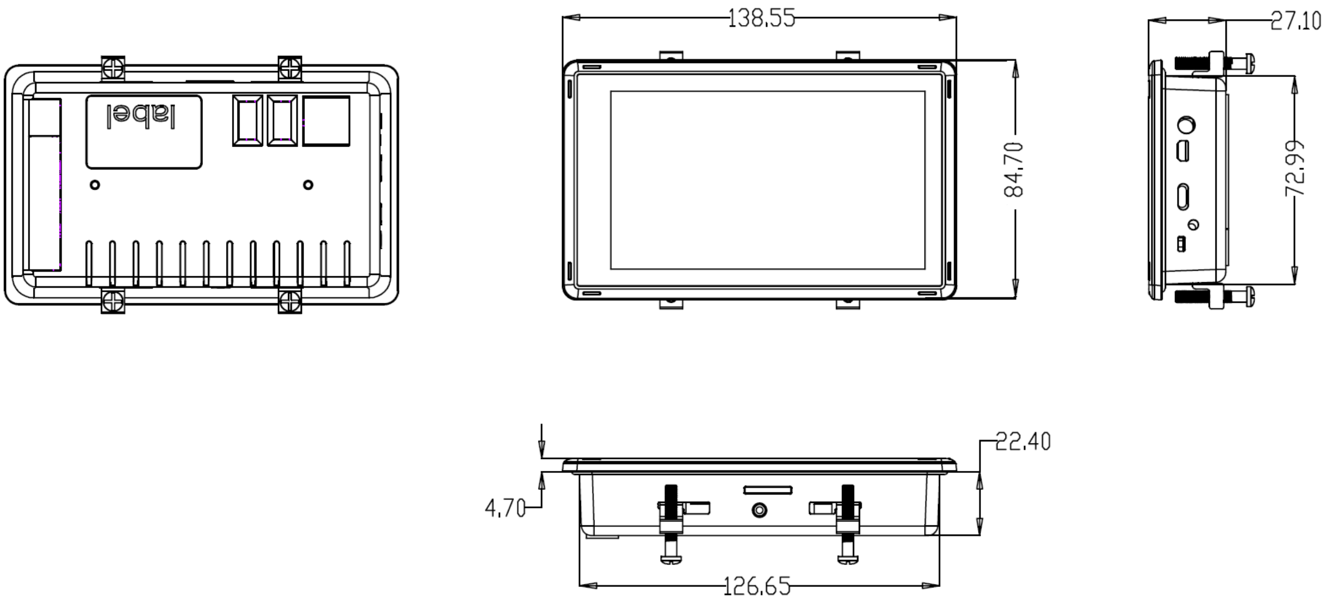
You can mount PPC-RK3576-050 with PANEL mounting (guide).

Attention

Please make sure the display is not exposed to high pressure when mounting into an enclosure.

Mechanical Specifications

For PPC-RK3576-050, the outer mechanical dimensions are 138.55 x 84.70 x 27.10mm (W x L x H).



Technical Drawing (PPC-RK3576-050)

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